Transformer-Based Spectral Unmixing for Lunar Mineral Mapping

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Abstract. Lunar mineral mapping remains challenging because of spectral dimensionality, mixing, nonlinear variability, and long-range dependencies. This study presents a one-dimensional Vision Transformer (ViT) for spectral unmixing and mineral abundance estimation using Chandrayaan-2 Imaging Infrared Spectrometer (IIRS) observations. Each spectrum is represented as spectral patches, and self-attention learns long-range relationships. The dual-head architecture reconstructs spectra and predicts non-negative, sum-to-one mineral fractions through a Softmax abundance head; chunk-wise inference enables full-scene mapping. On held-out IIRS observations, the model achieved an RMSE of 0.03637, MAE of 0.01420, and SAD of 0.02314 for spectral reconstruction, alongside an abundance-sum mean of 1.00000 and zero negative-abundance ratio. Full-scene inference produced predicted mineral distributions, with mare basalt, pyroxene, and spinel accounting for 63.61%, 16.90%, and 14.92% of mapped pixels, respectively. These results demonstrate the potential of one-dimensional transformer-based spectral learning for lunar mineral mapping.

Keywords: Hyperspectral imaging · Vision Transformer · Spectral unmixing · Lunar mineral mapping · Chandrayaan-2 IIRS · Mineral abundance estimation

1 Introduction

Hyperspectral imaging (HSI) is an important remote sensing technology for planetary surface analysis, providing continuous spectral information across hundreds of wavelength bands. In lunar exploration, HSI supports the characterization of surface mineralogy and geological composition. The Imaging Infrared Spectrometer (IIRS) onboard Chandrayaan-2 acquires 256 spectral bands spanning approximately 0.8–5.0 µm, providing diagnostic spectral information for lunar mineral identification [1,2].

Despite this capability, lunar hyperspectral analysis remains challenging owing to high spectral dimensionality, spectral redundancy, instrumental noise, illumination variability, and spectral mixing. Deep learning has improved hyperspectral representation learning through nonlinear feature extraction [11,7].

However, CNN-based methods primarily capture local spectral patterns and may have limited capability to model long-range dependencies across spectral bands.

Transformer-based methods address this limitation through self-attention and global spectral representation learning. SpectralFormer, spatial-spectral transformers, and dual attention transformers have demonstrated the effectiveness of transformer-based learning for hyperspectral analysis [8,9,12]. Vision Transformers (ViTs) represent inputs as token sequences processed through multi-head self-attention [3]. Nevertheless, most existing methods primarily address supervised hyperspectral classification, with comparatively limited exploration of transformer-based spectral unmixing for quantitative mineral abundance estimation.

An important distinction exists between spectral reconstruction and mineral abundance estimation. Reconstruction measures how closely predicted spectra approximate observed reflectance, whereas abundance estimation determines the fractional contribution of individual mineral components. Consequently, low reconstruction error alone does not establish geologically reliable mineral abundance estimates.

To address these limitations, this study proposes a one-dimensional Vision Transformer framework for spectral unmixing and lunar mineral mapping using Chandrayaan-2 IIRS observations. Each 256-band spectrum is divided into contiguous 16-band spectral patches and projected onto a token sequence. A pre-layer normalization ViT with six encoder layers, 256-dimensional embeddings, and eight attention heads models local spectral characteristics and long-range dependencies [3,4]. The shared latent representation supports two complementary heads for spectral reconstruction and mineral abundance prediction, with the latter subject to non-negativity and sum-to-one constraints.

The major contributions of this work are summarized as follows:

1. **A one-dimensional Vision Transformer framework** was proposed to learn long-range spectral dependencies from Chandrayaan-2 IIRS observations using spectral patch tokenization.
2. **A dual-head learning strategy** jointly performs spectral reconstruction and physically constrained mineral abundance prediction using a shared spectral representation.
3. **A scalable full-scene inference pipeline** for lunar mineral mapping was developed using chunk-wise processing, abundance reconstruction, Gaussian smoothing, and dominant-mineral assignment.
4. **An experimental evaluation** was performed on real IIRS observations using spectral reconstruction metrics, abundance consistency measures, and full-scene mineral distribution analysis.

2 Related Work

2.1 Hyperspectral Analysis and Spectral Unmixing

Hyperspectral analysis commonly employs spectral unmixing to decompose mixed pixels into constituent endmembers and their fractional abundances. Fully Con-

Table 1: Comparison of representative hyperspectral analysis and spectral unmixing approaches.

Approach	Endmember	Abundance	Recon.	Supervision	Complexity	Task
FCLSU [5]	Given	Yes	Yes	Unsupervised	Iterative	Unmixing
NMF [5]	Joint	Yes	Yes	Unsupervised	Iterative	Unmixing
CNN-based HSI [7]	N/A	N/A	N/A	Supervised	Architecture-dependent	Classification
DAEN [13]	Joint	Yes	Yes	Unsupervised	Multi-stage	Unmixing
DCTNU [15]	Yes	Yes	Yes	Data-driven	Multi-module	Unmixing
Double-DIP [16]	Joint	Yes	Yes	Unsupervised	Multi-module	Unmixing
SpectralFormer [8]	N/A	N/A	N/A	Supervised	Transformer	Classification
Proposed method	**Predefined**	**Yes**	**Yes**	**To be specified**	**Transformer**	**Unmixing + mapping**

strained Least Squares Unmixing (FCLSU) enforces non-negativity and abundance-sum constraints, whereas Non-negative Matrix Factorization (NMF) estimates non-negative endmember and abundance matrices through iterative optimization. However, FCLSU depends on predefined endmembers and linear mixing assumptions, while NMF is sensitive to initialization and may converge to local optima [5,14]. Nonlinear unmixing approaches address complex material interactions but introduce additional modelling complexity [16].

2.2 Deep Learning and Transformer-Based Methods

More recent unmixing-oriented transformer architectures combine convolutional feature extraction and attention to estimate endmembers and abundances [15].

2.3 Deep Learning-Based Spectral Unmixing

Deep Autoencoder Networks (DAEN) combine stacked autoencoders and variational autoencoders for unsupervised endmember and abundance estimation [13]. Convolutional autoencoder-based methods further incorporate spatial context into spectral reconstruction [17]. Similarly, deep image prior-based approaches investigate blind unmixing under linear and nonlinear mixing assumptions [16].

2.4 Research Gap

The literature reveals three major limitations: (1) conventional unmixing methods depend on predefined mixing assumptions or iterative optimization; (2) most transformer-based HSI studies emphasize classification, while deep unmixing methods commonly employ autoencoder-based or spatial–spectral architectures; and (3) computationally scalable one-dimensional spectral unmixing for complete orbital observations remains relatively unexplored. Table 1 summarizes the methodological differences between representative approaches.

3 Methodology

The proposed framework performs hyperspectral representation learning and lunar mineral mapping using Chandrayaan-2 IIRS observations. As shown in

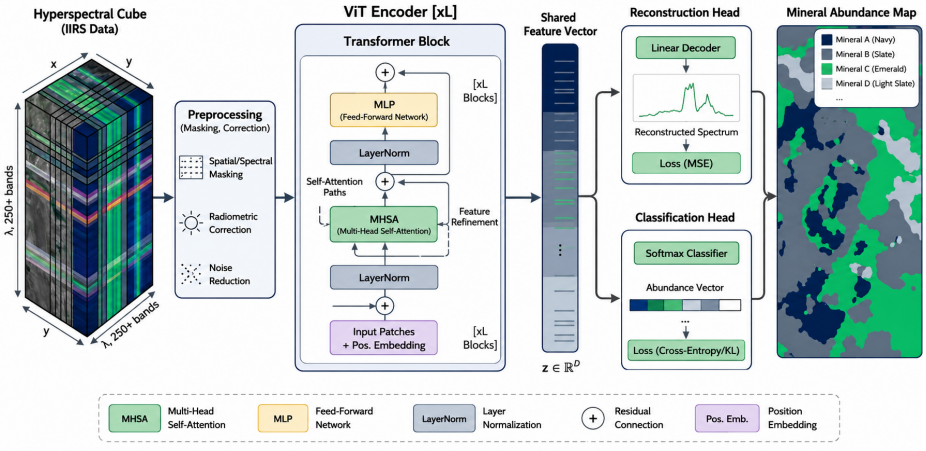


Fig. 1: Overall architecture of the proposed one-dimensional Vision Transformer for hyperspectral spectral unmixing and lunar mineral mapping.

Fig. 1, the complete pipeline comprises spectral preprocessing, one-dimensional spectral tokenization, Vision Transformer encoding, dual-task prediction, joint optimization, and scalable full-scene inference.

3.1 Data Preparation and Spectral Tokenisation

The Chandrayaan-2 IIRS observations contain 256 spectral channels spanning approximately 0.8–5.0 µm. Each spatial pixel is represented as a spectral signature $\mathbf{x} \in \mathbb{R}^{C_0}$, where $C_0 = 256$. Noisy edge channels and thermally contaminated wavelengths beyond 3.5 µm were masked, retaining C valid spectral channels. The observations were obtained from [VERIFY: exact IIRS data source and observation identifiers], covering [VERIFY: geographical coordinates or study region].

Photometric correction was applied during preprocessing to improve the radiometric consistency of IIRS observations [10]. The remaining spectra were independently normalized using min–max normalization:

$$x'_j = \frac{x_j - \min(\mathbf{x})}{\max(\mathbf{x}) - \min(\mathbf{x})}, \qquad j = 1, \ldots, C. \tag{1}$$

The normalized observations were divided into training, validation, and testing subsets using a [VERIFY: spatially disjoint or other] partitioning strategy, with a split ratio of [VERIFY: training/validation/testing percentages]. The corresponding sample counts were [VERIFY: training pixels], [VERIFY: validation pixels], and [VERIFY: testing pixels]. Spatial separation was maintained [VERIFY: explain the actual procedure used to prevent spatial leakage].

Each normalized spectrum is divided into non-overlapping spectral patches of length $P = 16$:

$$\tilde{\mathbf{x}}^{(i)} \in \mathbb{R}^P, \qquad N = \left\lfloor \frac{C}{P} \right\rfloor, \tag{2}$$

where N denotes the number of spectral tokens. For each patch, a linear projection maps the spectral values into a D-dimensional embedding space, followed by a learnable positional embedding:

$$\mathbf{z}^{(i)} = \mathbf{W}_E \tilde{\mathbf{x}}^{(i)} + \mathbf{e}^{(i)}. \tag{3}$$

A learnable class token is prepended to aggregate global spectral information:

$$\mathbf{Z}^{(0)} = [\mathbf{z}_{\text{cls}}, \mathbf{z}^{(1)}, \ldots, \mathbf{z}^{(N)}] \in \mathbb{R}^{(N+1)\times D}. \tag{4}$$

The mineral categories considered in this study comprise plagioclase, pyroxene, olivine, ilmenite, spinel, glass, anorthosite, and mare basalt. Their endmember/reference information was obtained from [VERIFY: actual spectral library or reference source], with the selection procedure described accordingly.

3.2 One-Dimensional Vision Transformer Encoder

The spectral token sequence is processed by a pre-layer normalization ViT consisting of L transformer blocks. Each block contains multi-head self-attention (MHSA) and a feed-forward network (FFN):

$$\mathbf{Z}'_\ell = \mathbf{Z}_{\ell-1} + \alpha\, \text{MHSA}\left(\text{LN}(\mathbf{Z}_{\ell-1})\right), \tag{5}$$

$$\mathbf{Z}_\ell = \mathbf{Z}'_\ell + \alpha\, \text{FFN}\left(\text{LN}(\mathbf{Z}'_\ell)\right). \tag{6}$$

Here, α denotes a learnable residual scaling parameter that adjusts the contribution of each attention and feed-forward branch to the residual representation. Pre-Layer Normalization improves optimization stability [4].

For attention head h,

$$\mathbf{Q}_h = \mathbf{Z}\mathbf{W}_h^Q, \qquad \mathbf{K}_h = \mathbf{Z}\mathbf{W}_h^K, \qquad \mathbf{V}_h = \mathbf{Z}\mathbf{W}_h^V, \tag{7}$$

and scaled dot-product attention is computed as

$$\text{Attention}_h = \text{softmax}\left(\frac{\mathbf{Q}_h \mathbf{K}_h^\top}{\sqrt{d_k}}\right)\mathbf{V}_h. \tag{8}$$

The FFN uses GELU activation [6]:

$$\text{FFN}(\mathbf{z}) = \mathbf{W}_2 \text{GELU}\left(\mathbf{W}_1 \mathbf{z} + \mathbf{b}_1\right) + \mathbf{b}_2. \tag{9}$$

The implemented encoder uses six transformer layers, a 256-dimensional embedding space, and eight attention heads. The training configuration comprises a batch size of [VERIFY], learning rate of [VERIFY], dropout rate of [VERIFY], weight decay of [VERIFY], and [VERIFY] training epochs. AdamW optimization with cosine learning-rate scheduling and linear warm-up is employed. Experiments were conducted using [VERIFY: processor/GPU, RAM, operating system, Python and PyTorch versions].

3.3 Dual-Task Learning and Optimization

The final class-token representation $\mathbf{z}_{\text{cls}}^{(L)} \in \mathbb{R}^D$ is shared by two prediction heads. The reconstruction head estimates the input spectrum:

$$\hat{\mathbf{x}} = \mathbf{W}_R \mathbf{z}_{\text{cls}}^{(L)} + \mathbf{b}_R. \tag{10}$$

The abundance head predicts the mineral contribution scores using Softmax:

$$\hat{\mathbf{a}} = \text{softmax}\left(\mathbf{W}_A \mathbf{z}_{\text{cls}}^{(L)} + \mathbf{b}_A\right). \tag{11}$$

For $K = 8$ mineral categories, the predicted abundance vector satisfies the non-negativity and sum-to-one constraints:

$$\sum_{k=1}^{K} \hat{a}_k = 1, \qquad \hat{a}_k \geq 0. \tag{12}$$

The joint optimization objective is expressed as

$$\mathcal{L} = \mathcal{L}_{\text{recon}} + \lambda \mathcal{L}_{\text{abund}}, \tag{13}$$

where $\mathcal{L}_{\text{recon}}$ denotes the spectral reconstruction loss, $\mathcal{L}_{\text{abund}}$ represents the abundance-learning objective, and λ controls their relative contribution. The reconstruction loss is calculated using mean squared error:

$$\mathcal{L}_{\text{recon}} = \frac{1}{BC} \sum_{i=1}^{B} \sum_{j=1}^{C} (x_{ij} - \hat{x}_{ij})^2, \tag{14}$$

where B represents the batch size and C denotes the number of retained spectral channels.

[VERIFY: Specify the actual abundance-learning objective, its reference targets, and the procedure used to generate or obtain those targets. If independent abundance labels are unavailable, clarify whether the objective uses pseudo-labels or an unsupervised regularization strategy.]

The two objectives are jointly optimized. Dropout is employed for regularization, and the model with the minimum validation loss is retained for evaluation.

3.4 Full-Scene Mineral Mapping

For complete orbital observations, the trained model is applied using chunk-wise inference to control memory consumption. Each scene is partitioned into spatial chunks of [VERIFY: chunk height × chunk width] and processed in mini-batches of [VERIFY: inference batch size]. Boundary regions are handled using [VERIFY: actual boundary handling procedure] to preserve the original spatial arrangement.

The predicted abundance vectors are reconstructed in their original spatial arrangement. Gaussian smoothing is subsequently applied independently to each

Algorithm 1 Full-Scene Lunar Mineral Mapping

Require: IIRS hyperspectral cube $\mathbf{X}$, trained model f_θ, chunk size C_L, batch size B_s
Ensure: Mineral map $\mathcal{M}$ and abundance maps $\hat{\mathbf{A}}$
Partition $\mathbf{X}$ into spatial chunks of C_L scan lines
Initialize abundance map $\hat{\mathbf{A}}$
for each spatial chunk $\mathbf{X}_c$ **do**
    Extract valid pixel spectra from $\mathbf{X}_c$
    Normalize spectra using (1)
    for each mini-batch $\mathbf{X}_b$ **do**
        $\hat{\mathbf{A}}_b \leftarrow f_\theta(\mathbf{X}_b)$
    end for
    Reconstruct spatial abundance arrangement for $\mathbf{X}_c$
    Insert $\hat{\mathbf{A}}_c$ into $\hat{\mathbf{A}}$
end for
Apply Gaussian smoothing to each abundance channel
for each spatial position (u, v) **do**
    $\mathcal{M}(u, v) \leftarrow \arg\max_k \hat{a}_k(u, v)$
end for
return $\hat{\mathbf{A}}, \mathcal{M}$

abundance channel using a kernel size of [VERIFY] and standard deviation of [VERIFY]. The dominant mineral at spatial position (u, v) is determined from the predicted abundance channels as

$$\mathcal{M}(u, v) = \arg\max_k \hat{a}_k(u, v). \tag{15}$$

The eight output channels are associated with the predefined mineral categories in the order specified during model configuration. Spatial continuity is examined through [VERIFY: actual spatial validation procedure or qualitative visual inspection].

4 Results and Discussion

4.1 Spectral Reconstruction

Figure 2 shows the representative best, median, and worst spectral reconstruction cases. The reconstructed spectra closely follow the corresponding IIRS observations and preserve the major absorption characteristics near approximately 1.0 and 2.0 µm.

The quantitative reconstruction results are presented in Table 2. The model achieved an RMSE of 0.03637, MAE of 0.01420, and SAD of 0.02314, indicating that the reconstructed spectra retain the dominant characteristics of the input observations. The corresponding SRE and ERGAS values are 28.78434 and 31.14898, respectively. These metrics assess spectral reconstruction quality rather than the accuracy of the predicted mineral abundances. Furthermore, the present results do not establish superiority over conventional or learning-based

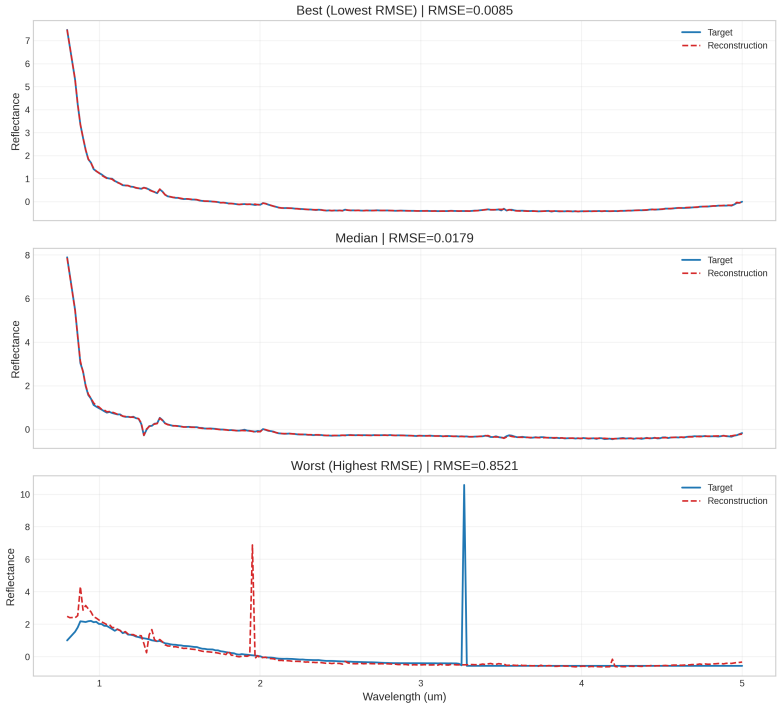


Fig. 2: Representative spectral reconstruction results for the best, median, and worst cases obtained by the proposed ViT.

unmixing methods; comparative evaluation using the same data and metrics is required to support such a conclusion.

The learned latent representations shown in Fig. 3 exhibit different patterns across the principal mineral categories. However, latent features should not be interpreted as physically measured mineral endmember signatures or independent evidence of mineralogical separability.

4.2 Mineral Abundance Estimation

The predicted abundance distributions are shown in Fig. 4a. Unlike hard classification, the proposed framework produces continuous mineral fractions for each spectral observation. This mathematical consistency is reflected by the abundance-sum mean of 1.00000 and standard deviation of 4.35×10^{-8} reported in Table 2.

The mean abundance distribution in Fig. 4b shows variation among the predicted mineral classes, with mare basalt and spinel representing prominent components of the evaluated observations.

The zero negative-abundance ratio reported in Table 2 confirms that the predictions satisfy the non-negativity constraint. Independent reference abun-

Table 2: Quantitative evaluation of the proposed ViT on the held-out IIRS test set.

Metric	**Proposed ViT**
RMSE	0.03637
MAE	0.01420
SAD	0.02314
SRE	28.78434
ERGAS	31.14898
PSNR (dB)	28.78434
Abundance Sum Mean	1.00000
Abundance Sum Std.	4.35×10^{-8}
Abundance L1 Mean	0.12500
Negative Abundance Ratio	0.00000
Abundance Entropy Mean	1.95543

Table 3: Mineral-class distribution in the evaluated full-scene mineralogical map.

Mineral Class	**Pixel Count**	**Percentage (%)**
Plagioclase	43,948	3.19
Pyroxene	232,881	16.90
Olivine	6,797	0.49
Ilmenite	8,642	0.63
Spinel	205,677	14.92
Glass	0	0.00
Anorthosite	3,566	0.26
Mare Basalt	876,739	63.61
Total	**1,378,250**	**100.00**

dances or mineralogical measurements are required to quantitatively validate the predicted fractions.

4.3 Full-Scene Mineral Mapping

The trained model was applied to the complete Chandrayaan-2 IIRS observations using the chunk-wise inference strategy described in Section 3. The resulting predicted mineral map is shown in Fig. 5.

The generated map exhibits spatially continuous mineral distributions with reduced isolated pixel-level variations. However, spatial continuity alone does not confirm geological correctness, and the mapped patterns require comparison with independent lunar mineralogical references.

The scene-level distribution is presented in Table 3. Mare basalt constituted the largest mapped fraction (63.61%), followed by pyroxene (16.90%) and

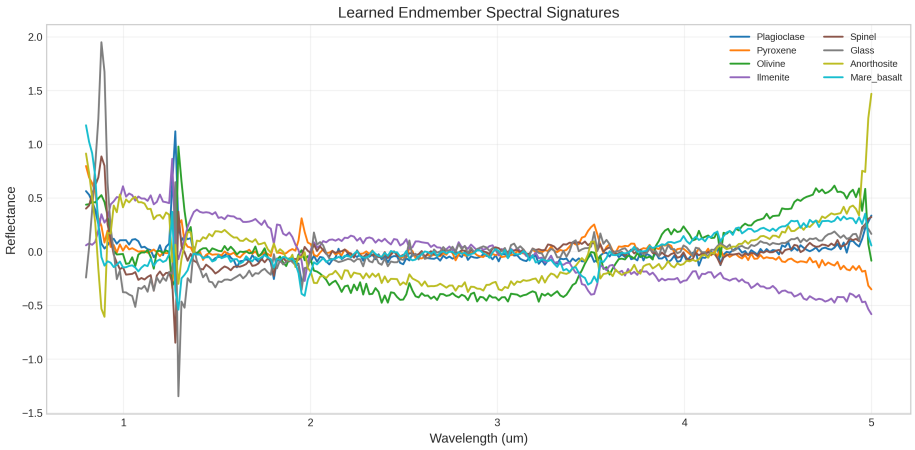
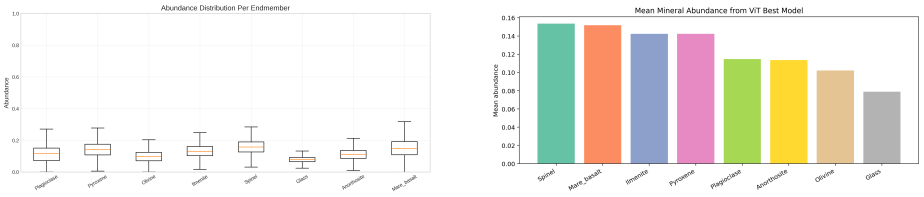


Fig. 3: Learned latent spectral feature representations associated with the principal lunar mineral categories extracted by the proposed ViT encoder.



(a) Predicted abundance distributions. (b) Mean predicted mineral abundance.

Fig. 4: Predicted mineral abundance distributions obtained by the proposed ViT.

spinel (14.92%). Plagioclase accounted for 3.19%, whereas olivine, ilmenite, and anorthosite represented smaller fractions. No pixels were assigned to the glass class in the evaluated map. These percentages represent the distribution of predicted dominant mineral labels and do not indicate independently measured mineral abundances.

5 Conclusion

This paper presents a one-dimensional Vision Transformer framework for hyperspectral unmixing and lunar mineral mapping using Chandrayaan-2 IIRS observations. By representing spectra as spectral patches, the model uses self-attention to capture long-range spectral dependencies and jointly performs spectral reconstruction and physically constrained abundance prediction through a dual-head architecture.

Current limitations include predefined mineral endmembers, variability in degraded orbital observations, and reliance on a single mission source. Future

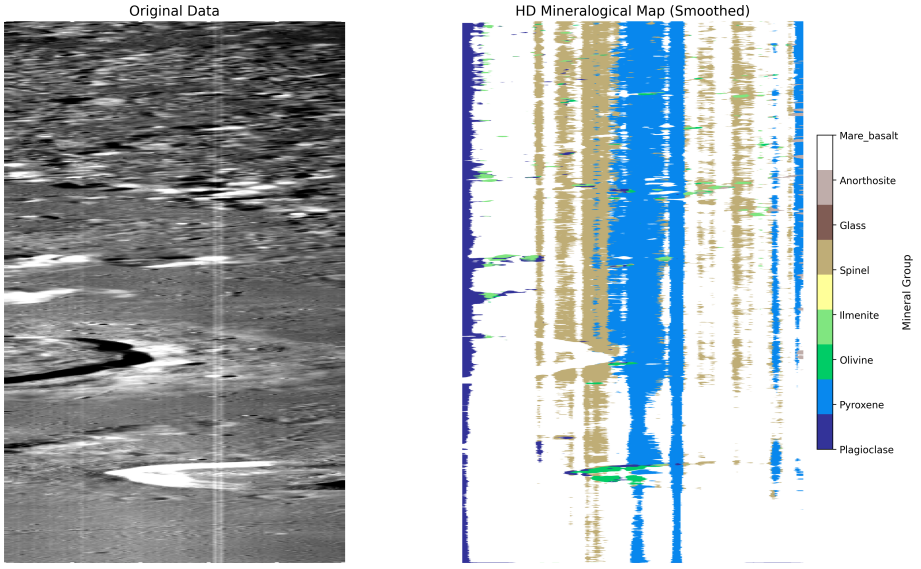


Fig. 5: Original Chandrayaan-2 IIRS observation and corresponding predicted mineral map generated by the proposed ViT.

work will explore adaptive endmember extraction, independent mineralogical validation, cross-scene and cross-mission generalization, and physics-informed self-supervised learning to improve the reliability and robustness of planetary hyperspectral analysis.

References

1. Chowdhury, A.R., Chauhan, P., Bugiolacchi, R., et al.: Planetary exploration of the Moon with Chandrayaan-2 Imaging Infrared Spectrometer (IIRS). Current Science **118**(3), 368–375 (2020). https://doi.org/10.18520/cs/v118/i3/368-375
2. Chauhan, P., et al.: Unambiguous detection of OH and H_2O on the Moon from Chandrayaan-2 Imaging Infrared Spectrometer reflectance data using the 3 μm hydration feature. Current Science **121**(3), 391–401 (2021). https://doi.org/10.18520/cs/v121/i3/391-401
3. Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., Uszkoreit, J., Houlsby, N.: An Image is Worth 16×16 Words: Transformers for Image Recognition at Scale. In: International Conference on Learning Representations (ICLR) (2021). https://doi.org/10.48550/arXiv.2010.11929
4. Xiong, R., Yang, Y., He, D., Zheng, K., Zheng, S., Xing, C., Zhang, H., Lan, Y., Wang, L., Liu, T.: On Layer Normalization in the Transformer Architecture. In: International Conference on Machine Learning (ICML), Proceedings of Machine Learning Research, vol. 119, pp. 10524–10533 (2020). https://doi.org/10.48550/arXiv.2002.04745

5. Bioucas-Dias, J.M., Plaza, A., Dobigeon, N., Parente, M., Du, Q., Gader, P., Chanussot, J.: Hyperspectral Unmixing Overview: Geometrical, Statistical, and Sparse Regression-Based Approaches. IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing **5**(2), 354–379 (2012). https://doi.org/10.1109/JSTARS.2012.2194696
6. Hendrycks, D., Gimpel, K.: Gaussian Error Linear Units (GELUs). arXiv preprint arXiv:1606.08415 (2016). https://doi.org/10.48550/arXiv.1606.08415
7. Paoletti, M.E., Haut, J.M., Plaza, J., Plaza, A.: Deep Learning Classifiers for Hyperspectral Imaging: A Review. ISPRS Journal of Photogrammetry and Remote Sensing **158**, 279–317 (2019). https://doi.org/10.1016/j.isprsjprs.2019.09.006
8. Hong, D., Han, Z., Yao, J., Gao, L., Zhang, B., Plaza, A., Chanussot, J.: SpectralFormer: Rethinking Hyperspectral Image Classification with Transformers. IEEE Transactions on Geoscience and Remote Sensing **60**, 1–15 (2022). https://doi.org/10.1109/TGRS.2021.3130716
9. He, X., Chen, Y., Lin, Z.: Spatial–Spectral Transformer for Hyperspectral Image Classification. Remote Sensing **13**(3), 498 (2021). https://doi.org/10.3390/rs13030498
10. Sharma, V., Chauhan, P., Kumar, A.: Photometric Correction of Images Obtained from Chandrayaan-2 Imaging Infra-Red Spectrometer (IIRS) Data. Advances in Space Research **73**(4), 2105–2118 (2024). https://doi.org/10.1016/j.asr.2023.09.015
11. Zhang, L., Zhang, L., Du, B.: Deep Learning for Remote Sensing Data: A Technical Tutorial on the State of the Art. IEEE Geoscience and Remote Sensing Magazine **4**(2), 22–40 (2016). https://doi.org/10.1109/MGRS.2016.2540798
12. Shu, Z., Wang, Y., Yu, Z.: Dual Attention Transformer Network for Hyperspectral Image Classification. Engineering Applications of Artificial Intelligence **127**, 107351 (2024).
13. Su, Y., Li, J., Plaza, A., Marinoni, A., Gamba, P., Chakravortty, S.: DAEN: Deep Autoencoder Networks for Hyperspectral Unmixing. IEEE Transactions on Geoscience and Remote Sensing **57**(7), 4309–4321 (2019). https://doi.org/10.1109/TGRS.2018.2890633
14. Chen, J., Zhao, M., Wang, X., Richard, C., Rahardja, S.: Integration of Physics-Based and Data-Driven Models for Hyperspectral Image Unmixing: A Summary of Current Methods. IEEE Signal Processing Magazine **40**(2), 61–74 (2023). https://doi.org/10.1109/MSP.2022.3208987
15. Hadi, F., Yang, J., Farooque, G., Xiao, L.: Deep Convolutional Transformer Network for Hyperspectral Unmixing. European Journal of Remote Sensing **56**(1), Article 2268820 (2023). https://doi.org/10.1080/22797254.2023.2268820
16. Zhou, C., Rodrigues, M.R.D.: Hyperspectral Blind Unmixing Using a Double Deep Image Prior. IEEE Transactions on Neural Networks and Learning Systems **35**(11), 16478–16492 (2024). https://doi.org/10.1109/TNNLS.2023.3294714
17. Yu, Y., Ma, Y., Mei, X., Fan, F., Huang, J., Li, H.: Multi-Stage Convolutional Autoencoder Network for Hyperspectral Unmixing. International Journal of Applied Earth Observation and Geoinformation **113**, Article 102981 (2022). https://doi.org/10.1016/j.jag.2022.102981